



Introduction to Machine Learning (CS 436/536)

Lecture-8: Logistic Regression
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Credit Approval

Compute a “credit score”

age	33 years
salary	50,000
debt	27,500
years employed	1
years at residence	2
...	...

Approve for credit?

age	salary	debt	
x_1	x_2	x_3	...
w_1	w_2	w_3	...
1	10	-5	...

$$creditscore = w_1x_1 + w_2x_2 + w_3x_3 + \dots$$

A Simple Learning Model

- Input vector $\mathbf{x} = [x_1, x_2, \dots, x_d]^T$
- Compute a “credit score” by giving importance weights to the different inputs: $creditscore = \sum_{i=1}^d w_i x_i$
- Decision rule:
 - If $creditscore > threshold$: **Approve** credit (good credit score)
 - If $creditscore < threshold$: **Deny** credit (poor credit score)
- How to choose the importance weights w_i ?
 - input x_i is important in deciding credit approval $\Rightarrow$ large w_i
 - input x_i has a beneficial effect to credit $\Rightarrow w_i > 0$ (weighs positively)
 - input x_i has an adverse effect on credit $\Rightarrow w_i < 0$ (weighs negatively)₃

A Simple Learning Model

- Decision rule:

- If $creditscore > threshold$: **Approve** credit (good credit score) $\Rightarrow$ output **+1**
- If $creditscore < threshold$: **Deny** credit (poor credit score) $\Rightarrow$ output **-1**

- Can be written formally as:

$$h(\mathbf{x}) = \text{sign} \left(\left(\sum_{i=1}^d w_i x_i \right) - threshold \right)$$

Simplifying a little...

$$h(\mathbf{x}) = \text{sign} \left(\left(\sum_{i=1}^d w_i x_i \right) + w_0 1 \right)$$

w_0 is a “bias weight” which corresponds to the threshold: Approve if $\sum_{i=1}^d w_i x_i > w_0$

A Simple Learning Model

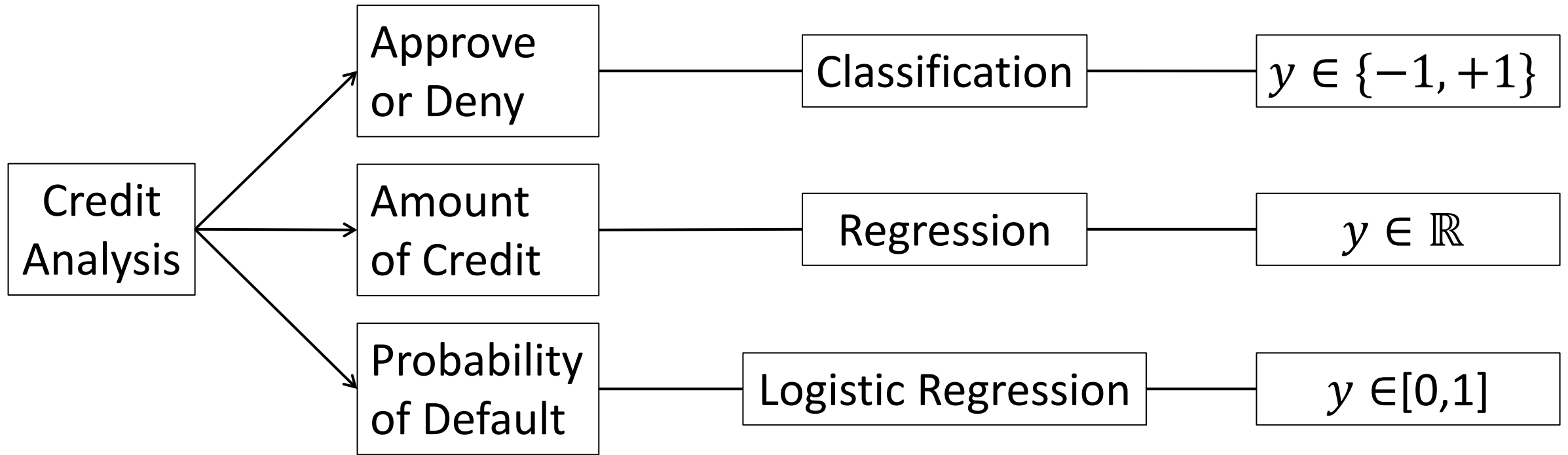
$$h(\mathbf{x}) = \text{sign} \left(\left(\sum_{i=1}^d w_i x_i \right) + w_0 1 \right)$$

$$= \text{sign}(w_0 1 + w_1 x_1 + w_2 x_2 + \cdots + w_d x_d)$$

$$\mathbf{w} = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ \vdots \\ w_d \end{bmatrix} \in \mathbb{R}^{d+1} \quad \mathbf{x} = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix} \in 1 \times \mathbb{R}^d \text{ (where } x_0 = 1)$$

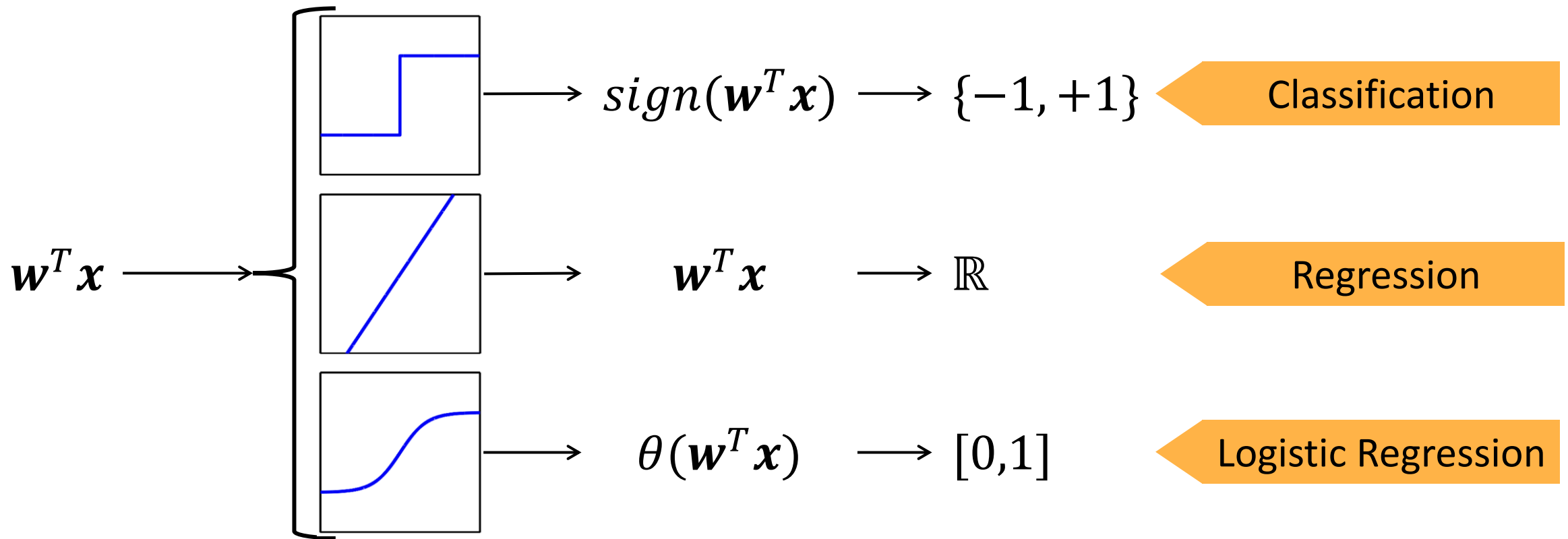
$$h(\mathbf{x}) = \text{sign}(\mathbf{w}^T \mathbf{x})$$

Linear Models for Three Learning Problems

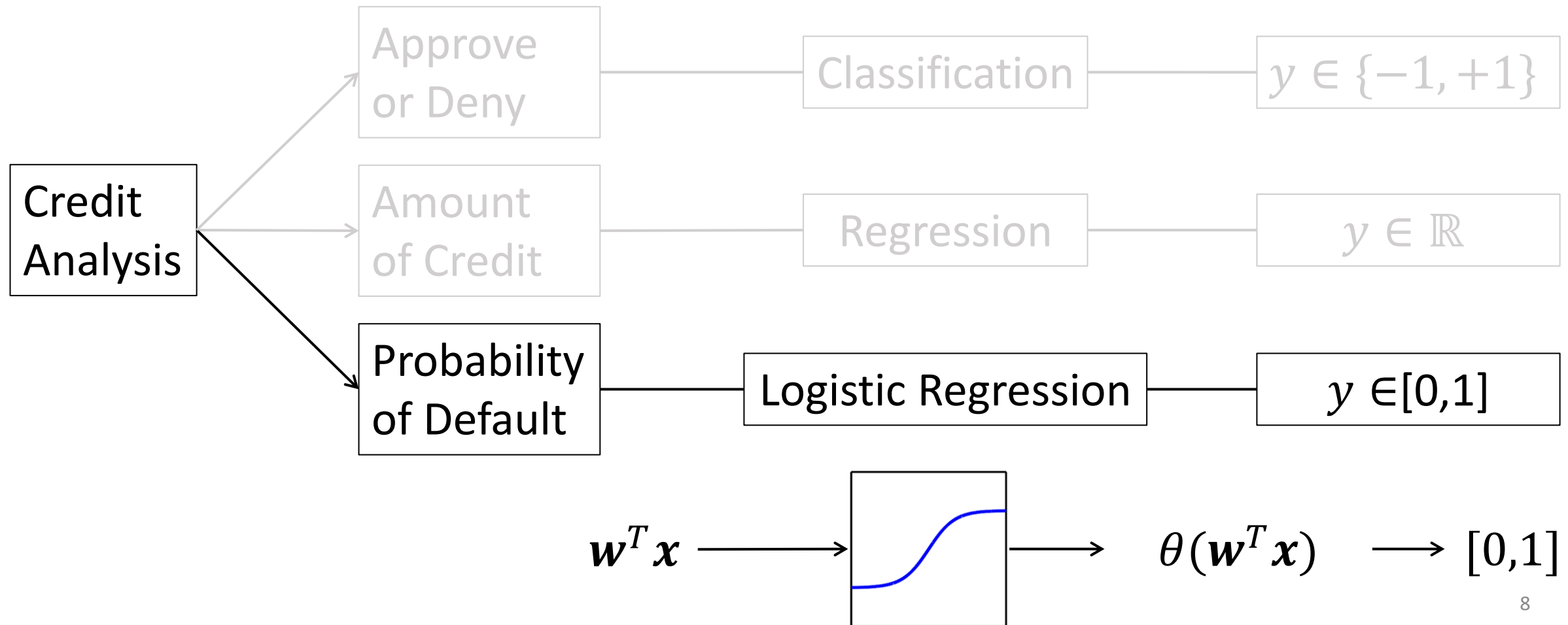


- Fundamental: Building block for more complex models
- First model to try!

Linear Models: The Linear Signal



Linear Models for Three Learning Problems



Predicting a Probability: Logistic Regression

Will the subject have a heart attack over the next year?

Patient Record

age	33 years
gender	male
blood pressure	137/91 mm Hg
HDL	50
LDL	120
BMI	23.7
...	...

Classification: Yes / No

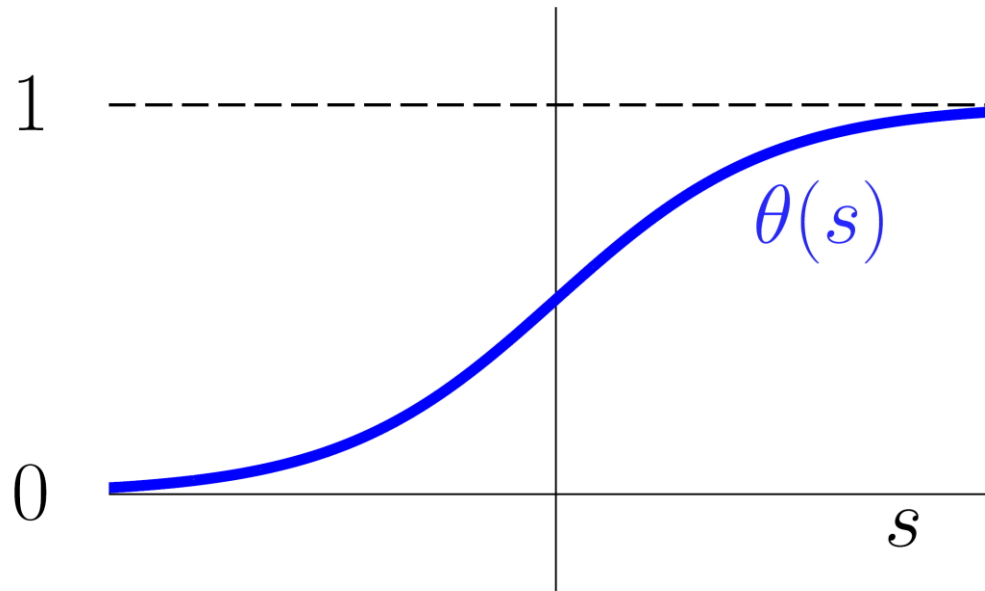
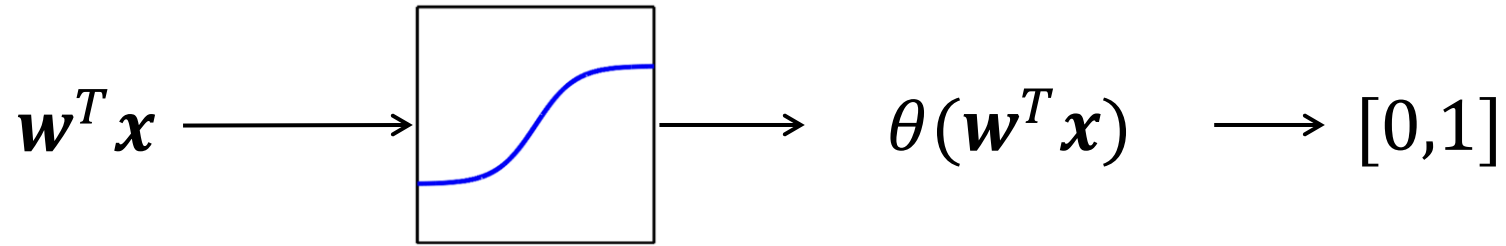
$$y \in \{-1, +1\}$$

Logistic Regression: Likelihood of heart attack

$$y \in [0,1]$$

$$h(\mathbf{x}) = \theta\left(\sum_{i=0}^d w_i x_i\right) = \theta(\mathbf{w}^T \mathbf{x}) \in [0,1]$$

Predicting a Probability Using the Linear Signal



$$\theta(s) = \frac{e^s}{1 + e^s} = \frac{1}{1 + e^{-s}}$$

$$\begin{aligned}\theta(-s) &= \frac{e^{-s}}{1 + e^{-s}} = \frac{1}{1 + e^s} \\ &= 1 - \theta(s)\end{aligned}$$

The Logistic Regression Setting: Data is Still Binary

$$\mathcal{D} = (\mathbf{x}_1, y_1 = \pm 1), \dots, (\mathbf{x}_N, y_N = \pm 1)$$

$\mathbf{x}_n$

Patient's health record

$y_n \in \{-1, +1\}$

Did they have a heart attack?

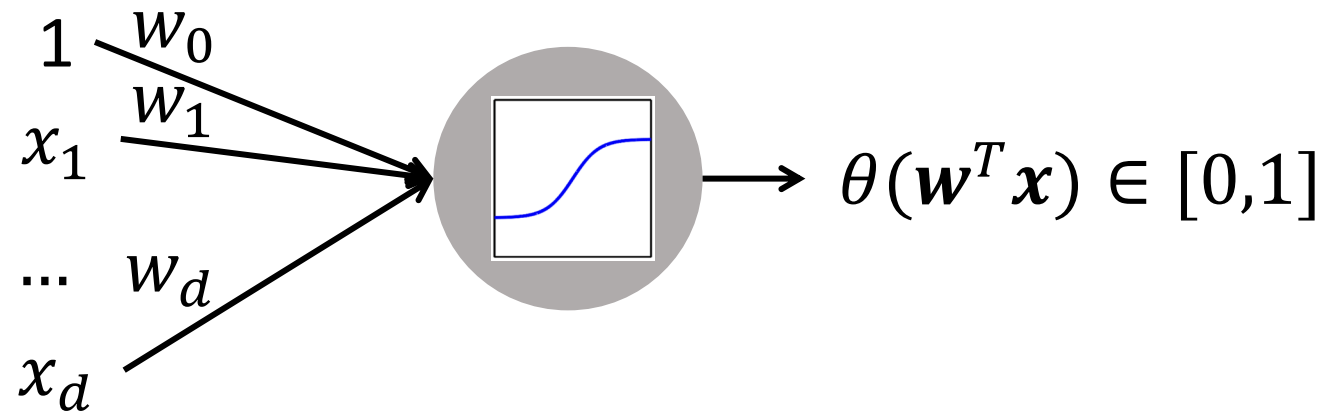
Cannot measure this as a probability

Event either occurs or does not

Logistic Regression:

- Given observations of an event
- Infer its probability

The Sigmoid Neuron



Predicting Categorical Targets Using Regression:

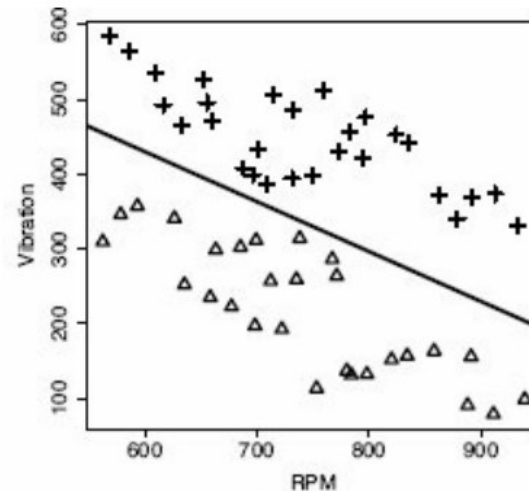
A dataset listing features for a number of generators.

ID	RPM	VIBRATION	STATUS
1	568	585	good
2	586	565	good
3	609	536	faulty

Predicting Categorical Targets Using Regression:

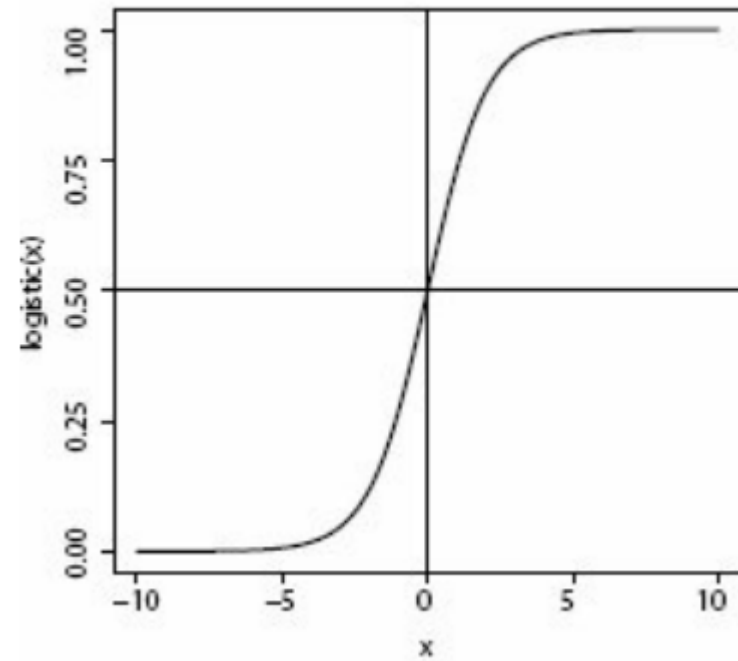
A dataset listing features for a number of generators.

ID	RPM	VIBRATION	STATUS
1	568	585	faulty
2	586	565	good
3	609	536	good



Logistic Function

$$\text{logistic}(x) = \frac{1}{1 + e^{-x}}$$



Logistic Regression Derivative:

$$\begin{aligned} & \frac{d}{dz} \frac{1}{1 + e^{-z}} \\ = & \frac{1}{(1 + e^{-z})^2} (e^{-z}) \\ = & \frac{1}{(1 + e^{-z})} \cdot \left(1 - \frac{1}{(1 + e^{-z})} \right) \end{aligned}$$

Notation Explanation:

$M_w(d) \Rightarrow$ Predicted Output from our model, logistic model in this case

$w \Rightarrow$ weights , $d \Rightarrow$ inputs

In case of logistic regression with binary classification it gives the probability. Therefore, we can say that:

$$P(t|d) = M_w(d)$$

$L_2(M_w, D) \Rightarrow$ Loss function as a function of predicted output and input.

$t \Rightarrow$ Actual Output

In coming examples W and B are interchangeably used to show weights:

$$M_w(d) = \frac{1}{1 + e^{-(w_0 + w_1 d_1 + w_2 d_2 + \dots + w_n d_n)}}$$

Logistic Regression

$$\begin{aligned}\mathbb{M}_{\mathbf{w}}(\mathbf{d}) &= \text{logistic}(\mathbf{w} \cdot \mathbf{d}) \\ &= \frac{1}{1 + e^{-\mathbf{w} \cdot \mathbf{d}}}\end{aligned}$$

$$P(t = \textit{faulty} | \mathbf{d}) = \mathbb{M}_{\mathbf{w}}(\mathbf{d})$$

$$P(t = \textit{good} | \mathbf{d}) = 1 - \mathbb{M}_{\mathbf{w}}(\mathbf{d})$$

Logistic Regression Learning with Gradient Descent

$$\frac{\partial}{\partial \mathbf{w}[j]} L_2(\mathbb{M}_{\mathbf{w}}, \mathcal{D}) = \frac{\partial}{\partial \mathbf{w}[j]} \frac{1}{2} (t - \mathbb{M}_{\mathbf{w}}(\mathbf{d}))^2$$

where $w[j]$ is a single weight from the set of weights $\mathbf{w}$. Applying the chain rule to this, we get

$$\frac{\partial}{\partial \mathbf{w}[j]} L_2(\mathbb{M}_{\mathbf{w}}, \mathcal{D}) = (t - \mathbb{M}_{\mathbf{w}}(\mathbf{d})) \times \frac{\partial}{\partial \mathbf{w}[j]} (t - \mathbb{M}_{\mathbf{w}}(\mathbf{d}))$$

But $\mathbb{M}_{\mathbf{w}}(\mathbf{d}) = \text{logistic}(\mathbf{w} \cdot \mathbf{d})$, so

$$\frac{\partial}{\partial \mathbf{w}[j]} L_2(\mathbb{M}_{\mathbf{w}}, \mathcal{D}) = (t - \text{logistic}(\mathbf{w} \cdot \mathbf{d})) \times \frac{\partial}{\partial \mathbf{w}[j]} (t - \text{logistic}(\mathbf{w} \cdot \mathbf{d}))$$

Final Update Rule:

$$\mathbf{w}[j] \leftarrow \mathbf{w}[j] + \alpha \times \sum_{i=1}^n ((t_i - \mathbb{M}_{\mathbf{w}}(\mathbf{d}_i)) \times \mathbb{M}_{\mathbf{w}}(\mathbf{d}_i) \times (1 - \mathbb{M}_{\mathbf{w}}(\mathbf{d}_i)) \times \mathbf{d}_i[j])$$

Final Update Rule:

$$\mathbf{w}[j] \leftarrow \mathbf{w}[j] + \alpha \times \sum_{i=1}^n ((t_i - \mathbb{M}_{\mathbf{w}}(\mathbf{d}_i)) \times \mathbb{M}_{\mathbf{w}}(\mathbf{d}_i) \times (1 - \mathbb{M}_{\mathbf{w}}(\mathbf{d}_i)) \times \mathbf{d}_i[j])$$

Working Example:

X1	X2	Y
2.7810836	2.550537003	0
1.465489372	2.362125076	0
3.396561688	4.400293529	0
1.38807019	1.850220317	0
3.06407232	3.005305973	0
7.627531214	2.759262235	1
5.332441248	2.088626775	1
6.922596716	1.77106367	1
8.675418651	-0.242068655	1
7.673756466	3.508563011	1

$$\text{output} = B_0 + B_1 \times X_1 + B_2 \times X_2$$

Working Example:

Initialization

$$B0 = 0.0$$

$$B1 = 0.0$$

$$B2 = 0.0$$

First Training Sample

$$prediction = \frac{1}{1 + e^{-(B0 + B1 \times X1 + B2 \times X2)}}$$

$$prediction = \frac{1}{1 + e^{-(0.0 + 0.0 \times 2.7810836 + 0.0 \times 2.550537003)}}$$

$$prediction = 0.5$$

Update rule for one sample

$$b = b + \alpha \times (y - \text{prediction}) \times \text{prediction} \times (1 - \text{prediction}) \times x$$

$$B0 = 0.0 + 0.3 \times (0 - 0.5) \times 0.5 \times (1 - 0.5) \times 1.0$$

$$B1 = B1 + 0.3 \times (0 - 0.5) \times 0.5 \times (1 - 0.5) \times 2.7810836$$

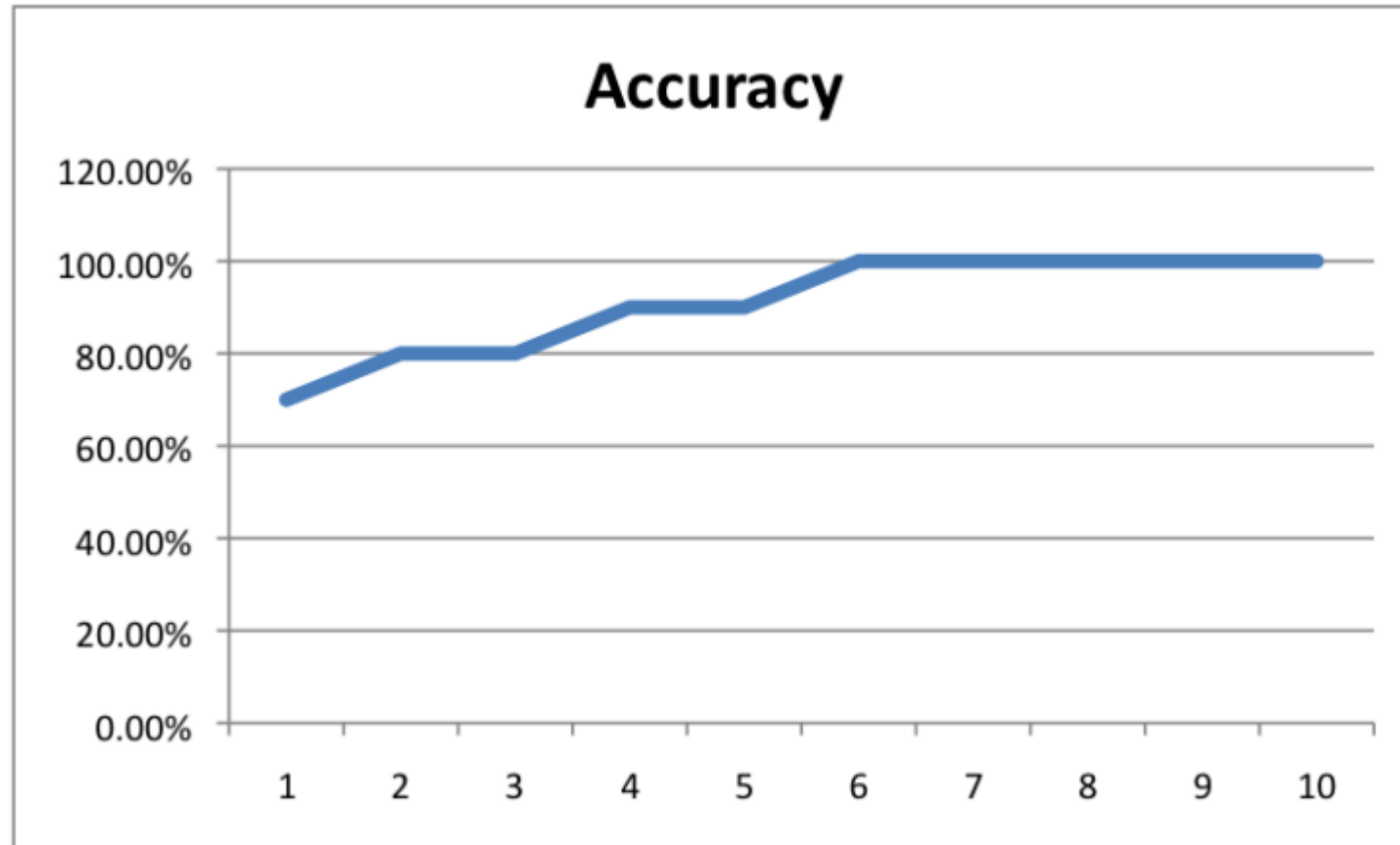
$$B2 = B2 + 0.3 \times (0 - 0.5) \times 0.5 \times (1 - 0.5) \times 2.550537003$$

$$B0 = -0.0375$$

$$B1 = -0.104290635$$

$$B2 = -0.095645138$$

If we continue for multiple iterations:



Final Convergence:

$$B0 = -0.406605464$$

$$B1 = 0.852573316$$

$$B2 = -1.104746259$$

Final Result:

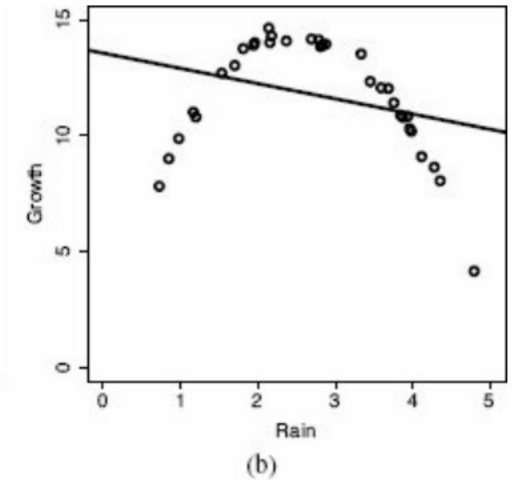
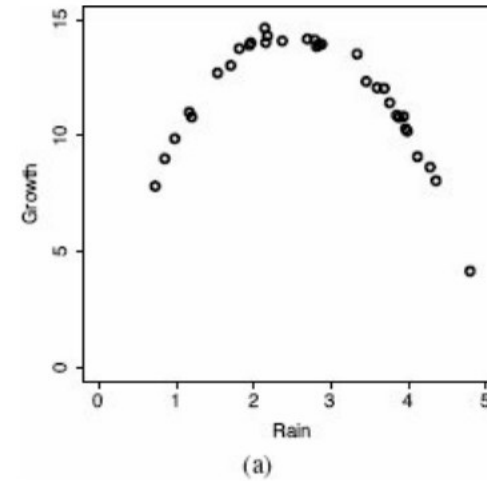
$prediction = \text{IF } (output < 0.5) \text{ Then } 0 \text{ Else } 1$

Prediction	Crisp
0.298756986	0
0.145951056	0
0.085333265	0
0.219737314	0
0.247059	0
0.954702135	1
0.862034191	1
0.971772905	1
0.999295452	1
0.905489323	1

$$accuracy = \frac{CorrectPredictions}{TotalPredictions} \times 100$$
$$accuracy = \frac{10}{10} \times 100$$
$$accuracy = 100\%$$

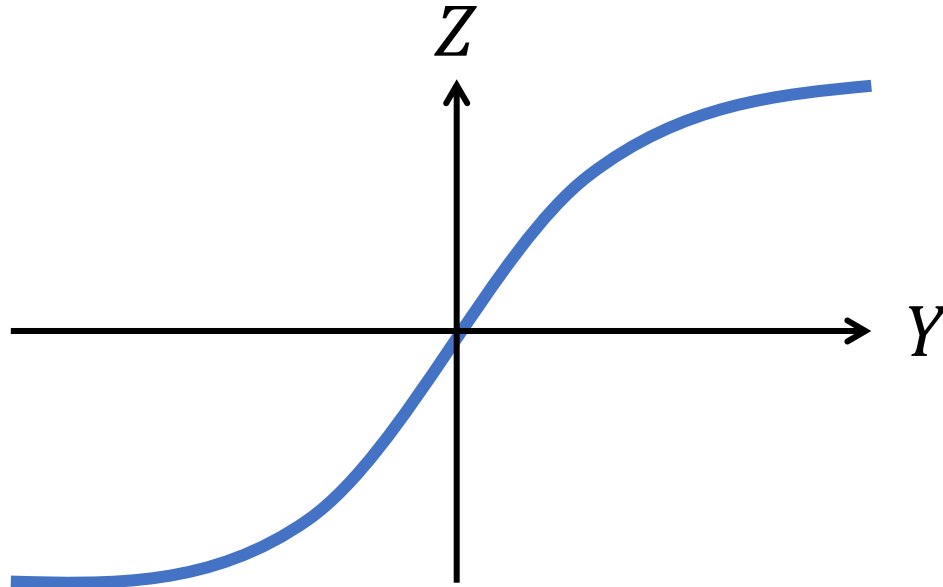
What if the relationship is non-linear between the variables?

ID	RAIN	GROWTH
1	2.153	14.016
2	3.933	10.834
3	1.699	13.026
4	1.164	11.019
5	4.793	4.162
6	2.690	14.167
7	3.982	10.190
8	3.333	13.525
9	1.942	13.899
10	2.876	13.949
11	4.277	8.643
12	3.754	11.420
13	2.809	13.847
14	1.809	13.757
15	4.114	9.101
16	2.834	13.923
17	3.872	10.795
18	2.174	14.307
19	4.353	8.059
20	3.684	12.041

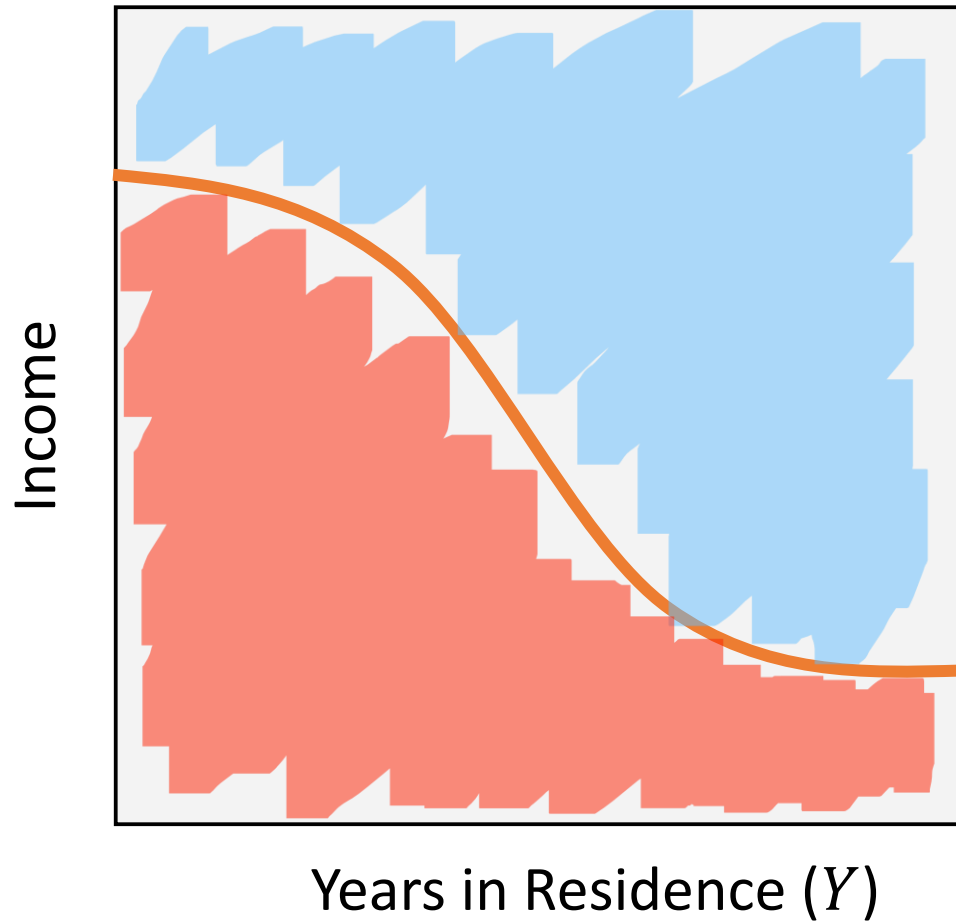


Change the Feature Using a Transform

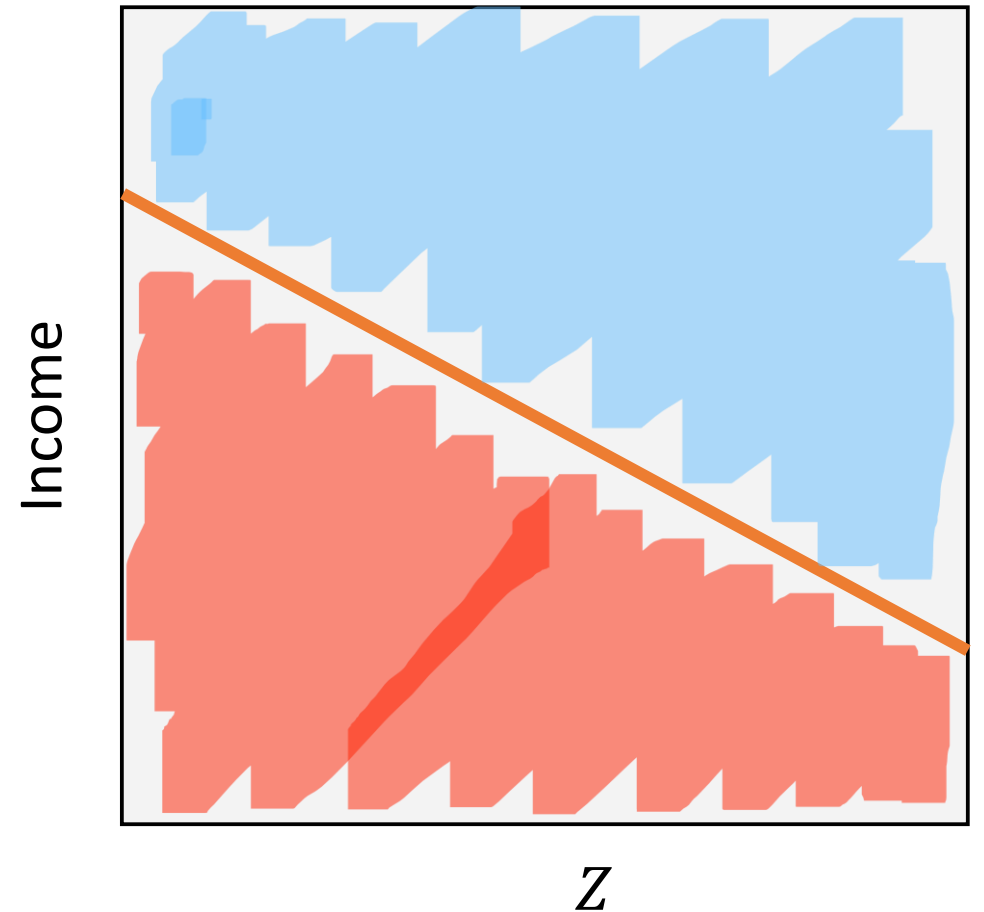
$$Y \xrightarrow{\Phi} Z$$



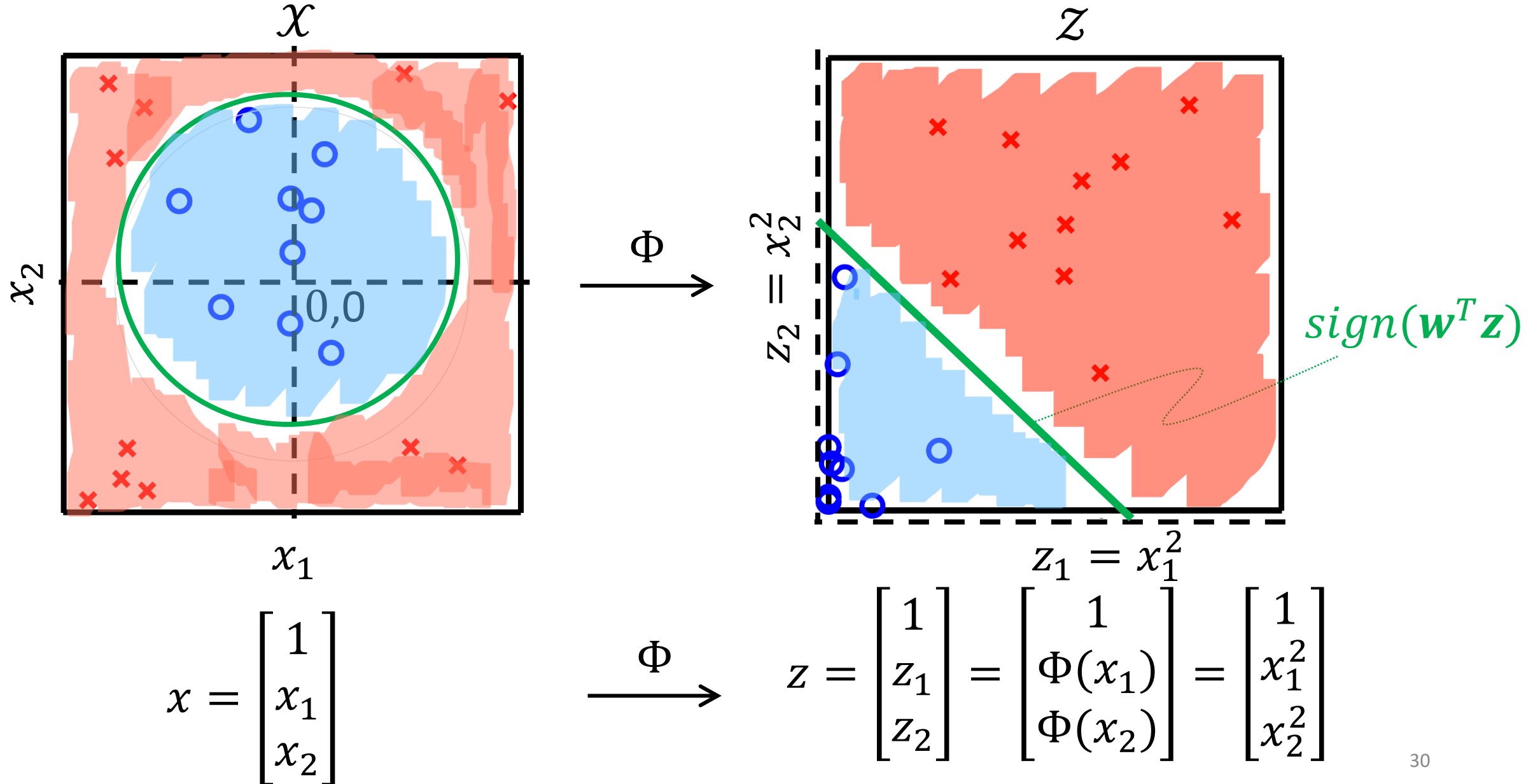
Changing the Feature Using a Transform



$$Y \xrightarrow{\Phi} Z$$



Mechanics of Non-Linear Feature Transforms



Solution?

$$\mathbb{M}_{\mathbf{w}}(\mathbf{d}) = \sum_{k=0}^b \mathbf{w}[k] \times \phi_k(\mathbf{d})$$

$$\text{GROWTH} = \mathbf{w}[0] \times \phi_0(\text{RAIN}) + \mathbf{w}[1] \times \phi_1(\text{RAIN}) + \mathbf{w}[2] \times \phi_2(\text{RAIN})$$

where

$$\phi_0(\text{RAIN}) = 1$$

$$\phi_1(\text{RAIN}) = \text{RAIN}$$

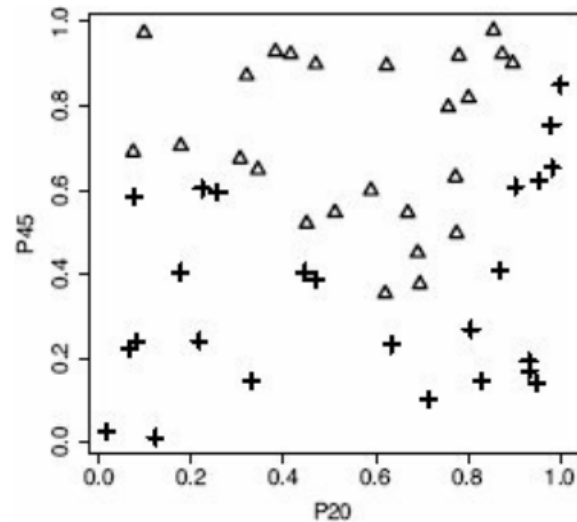
$$\phi_2(\text{RAIN}) = \text{RAIN}^2$$

$$\text{GROWTH} = 3.707 \times \phi_0(\text{RAIN}) + 8.475 \times \phi_1(\text{RAIN}) + -1.717 \times \phi_2(\text{RAIN})$$

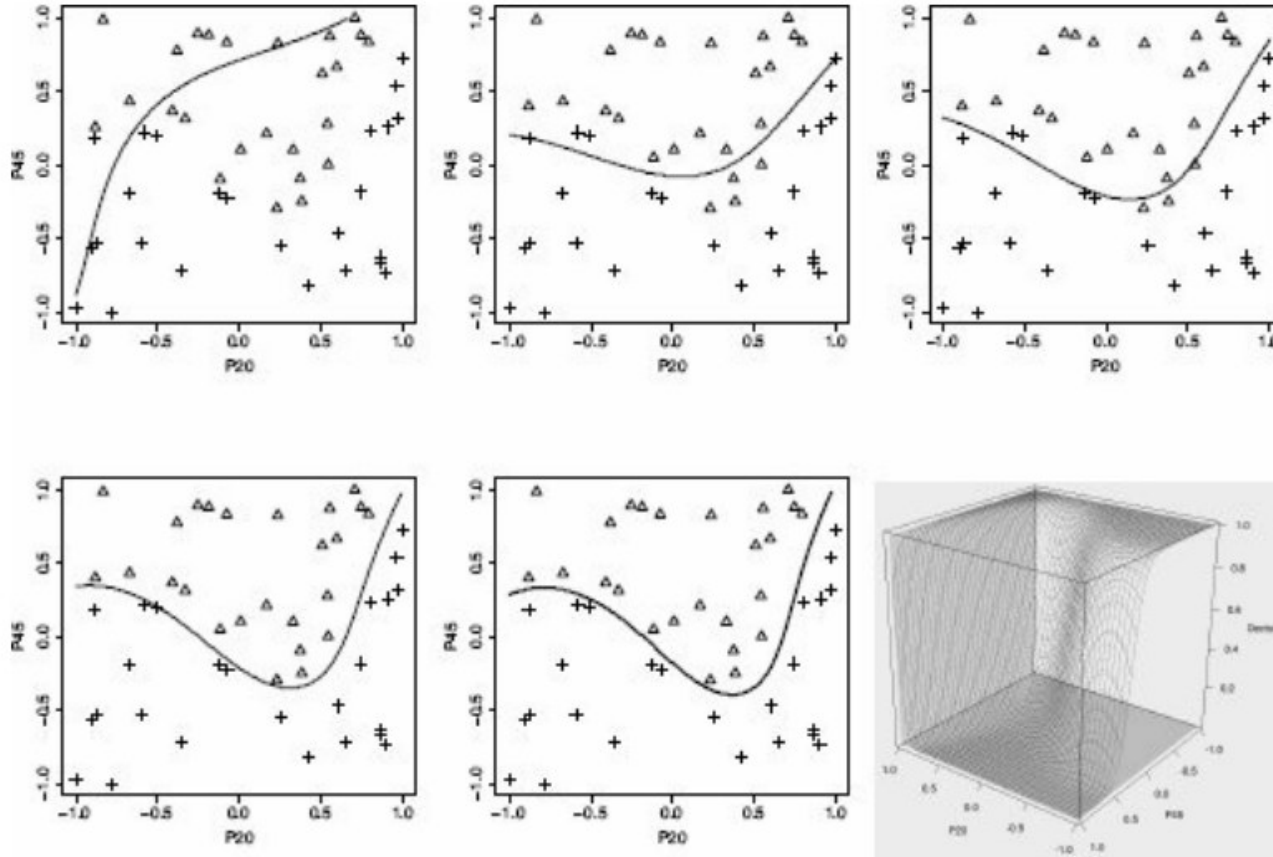
Multi-Variable Non-Linear Relationship:

P20	P45	TYPE
0.4497	0.4499	negative
0.8964	0.9006	negative
0.6952	0.3760	negative
0.1769	0.7050	negative
0.6904	0.4505	negative
0.7794	0.9190	negative

P20	P45	TYPE
0.0656	0.2244	positive
0.6336	0.2312	positive
0.4453	0.4052	positive
0.9998	0.8493	positive
0.9027	0.6080	positive
0.3319	0.1473	positive



One possible Hypothesis:



$$\mathbb{M}_{\mathbf{w}}(\mathbf{d}) = \frac{1}{1 + e^{-\left(\sum_{j=0}^b \mathbf{w}[j] \phi_j(\mathbf{d})\right)}}$$

$$\phi_0(\langle \mathbf{P20}, \mathbf{P45} \rangle) = 1$$

$$\phi_1(\langle \mathbf{P20}, \mathbf{P45} \rangle) = \mathbf{P20}$$

$$\phi_2(\langle \mathbf{P20}, \mathbf{P45} \rangle) = \mathbf{P45}$$

$$\phi_3(\langle \mathbf{P20}, \mathbf{P45} \rangle) = \mathbf{P20}^2$$

$$\phi_4(\langle \mathbf{P20}, \mathbf{P45} \rangle) = \mathbf{P45}^2$$

$$\phi_5(\langle \mathbf{P20}, \mathbf{P45} \rangle) = \mathbf{P20}^3$$

$$\phi_6(\langle \mathbf{P20}, \mathbf{P45} \rangle) = \mathbf{P45}^3$$

$$\phi_7(\langle \mathbf{P20}, \mathbf{P45} \rangle) = \mathbf{P20} \times \mathbf{P45}$$

Multiple Output Class:

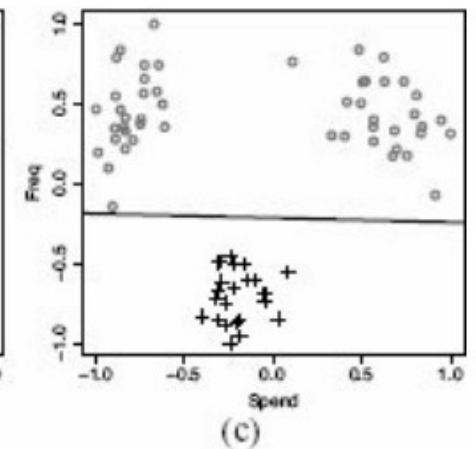
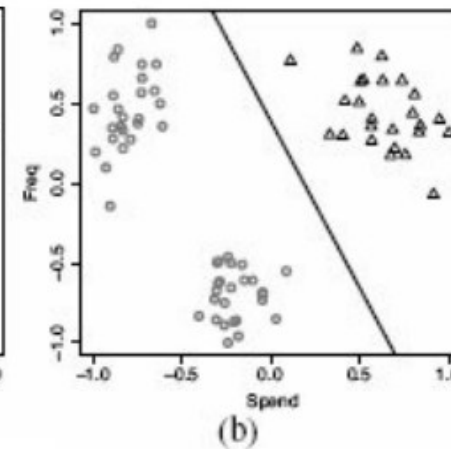
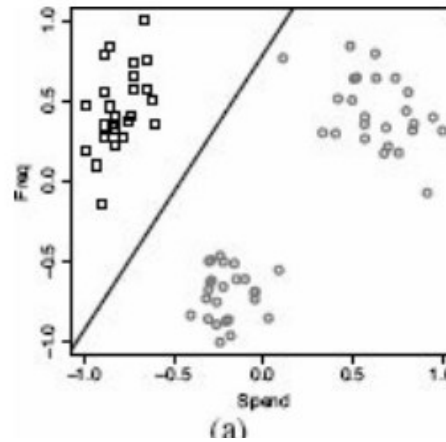
For r target feature levels, we build r separate logistic regression models M_{w_1} to M_{w_r} :

$$M_{w_1}(d) = \text{logistic}(w_1 \cdot d)$$

$$M_{w_2}(d) = \text{logistic}(w_2 \cdot d)$$

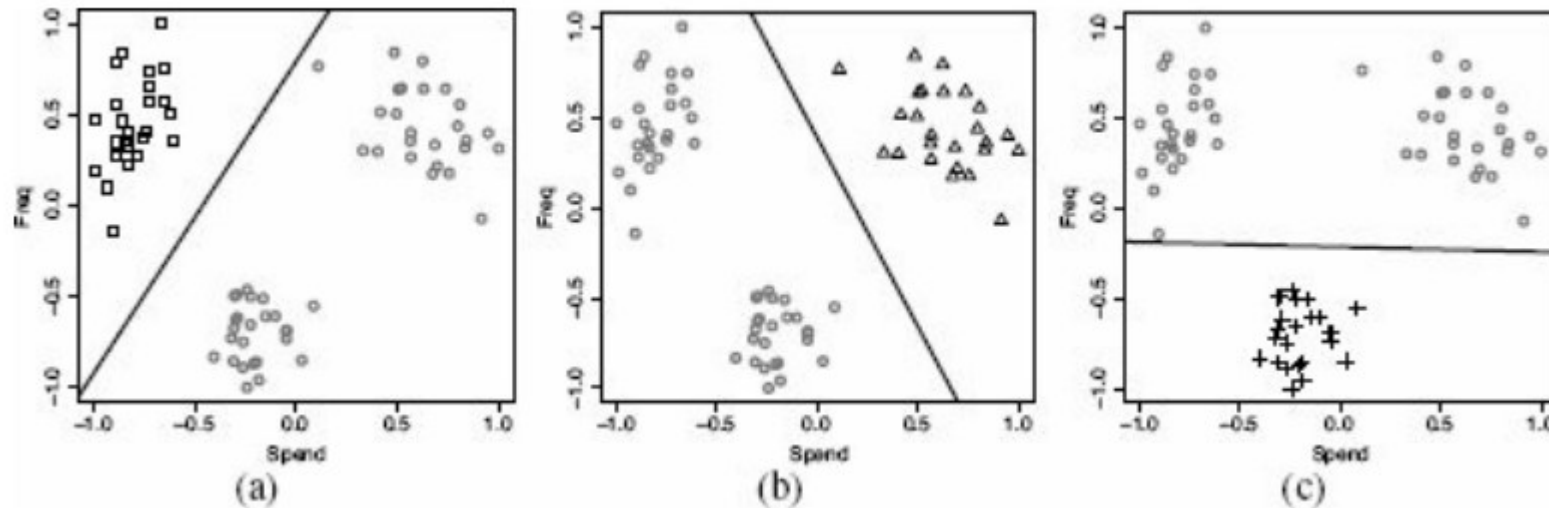
$\vdots$

$$M_{w_r}(d) = \text{logistic}(w_r \cdot d)$$



ID	SPEND	FREQ	TYPE	ID	SPEND	FREQ	TYPE
1	21.6	5.4	single	28	122.6	6.0	business
2	25.7	7.1	single	29	107.7	5.7	business
3	18.9	5.6	single			$\vdots$	
4	25.7	6.8	single	47	53.2	2.6	family
		$\vdots$		48	52.4	2.0	family
26	107.9	5.8	business	49	46.1	1.4	family
27	92.9	5.5	business	50	65.3	2.2	family
		$\vdots$				$\vdots$	

Multiple Output Class:



For r target feature levels, we build r separate logistic regression models $\mathbb{M}_{w_1}$ to $\mathbb{M}_{w_r}$:

$$\mathbb{M}_{w_1}(\mathbf{d}) = \text{logistic}(\mathbf{w}_1 \cdot \mathbf{d})$$

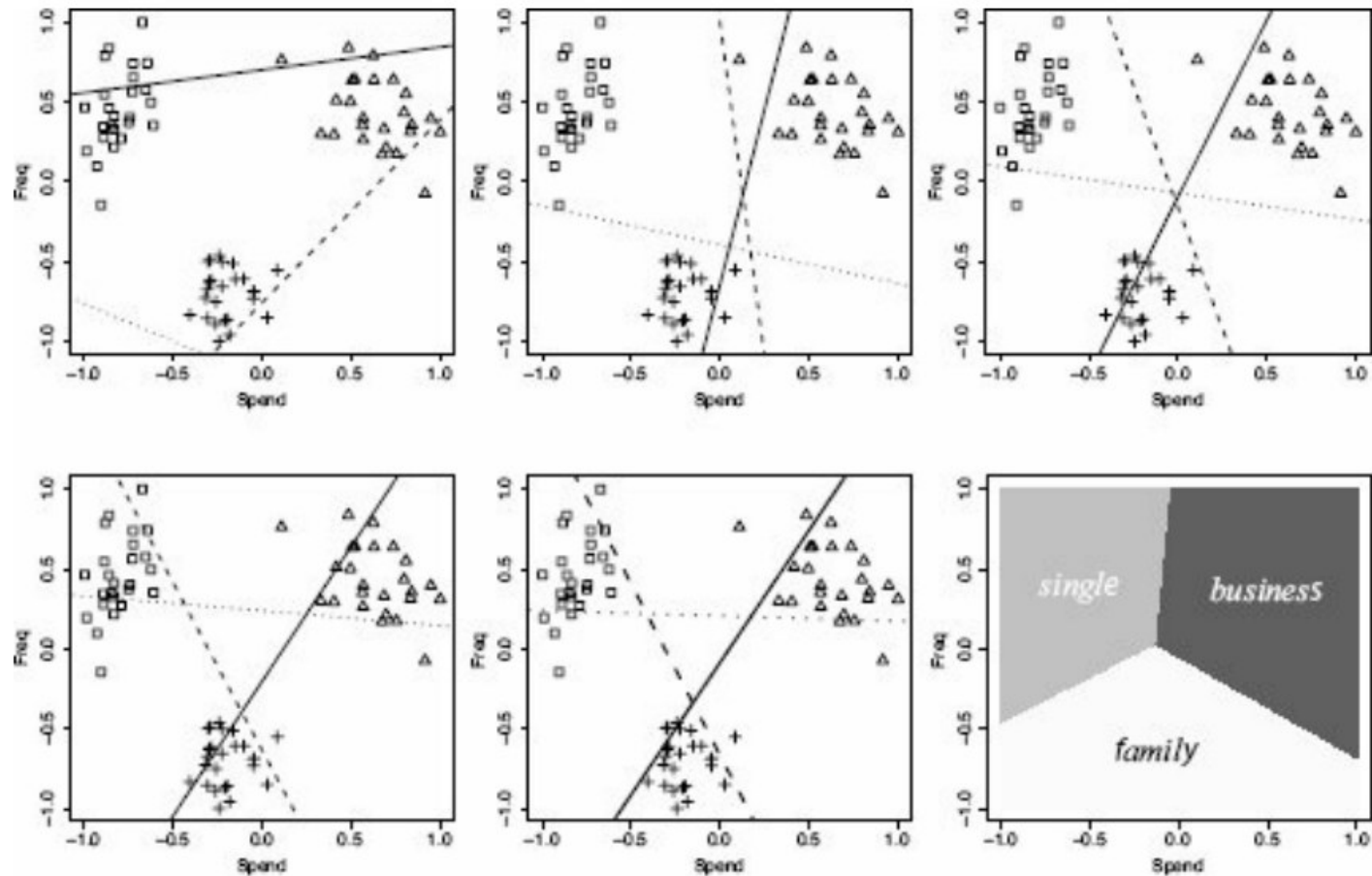
$$\mathbb{M}_{w_2}(\mathbf{d}) = \text{logistic}(\mathbf{w}_2 \cdot \mathbf{d})$$

$$\vdots$$

$$\mathbb{M}_{w_r}(\mathbf{d}) = \text{logistic}(\mathbf{w}_r \cdot \mathbf{d})$$

$$\mathbb{M}'_{w_k}(\mathbf{d}) = \frac{\mathbb{M}_{w_k}(\mathbf{d})}{\sum_{l \in \text{levels}(t)} \mathbb{M}_{w_l}(\mathbf{d})}$$

Multiple Output Class:



Problem with multiple output class:

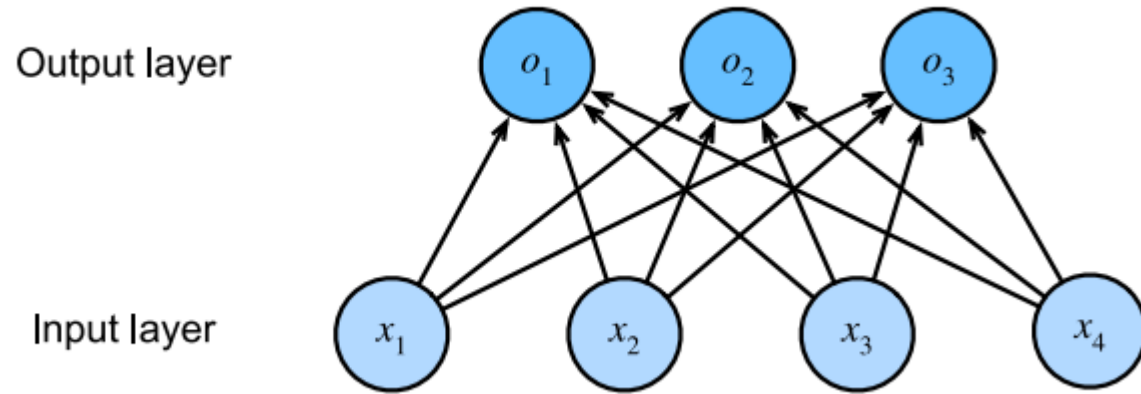
In practice, we are more often interested in *classification*: asking not “how much” but “which one”:

- Does this email belong in the spam folder or the inbox?
- Is this customer more likely *to sign up* or *not to sign up* for a subscription service?
- Does this image depict a donkey, a dog, a cat, or a rooster?
- Which movie is Aston most likely to watch next?

$$y \in \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}.$$

Let's Consider a case with three classes: a dog, a donkey and a cat.

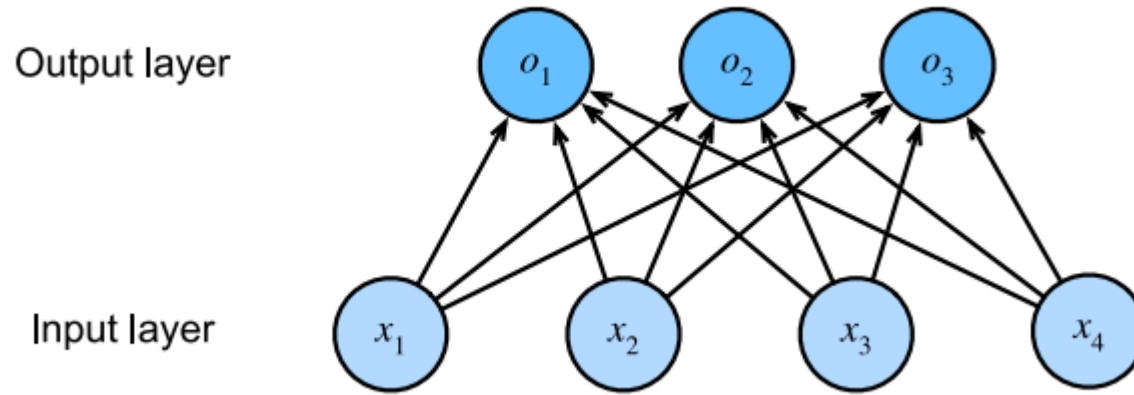
Solution: SoftMax Regression



$$\mathbf{o} = \mathbf{XW} + \mathbf{b},$$

$$\hat{\mathbf{Y}} = \text{softmax}(\mathbf{o}).$$

SoftMax Regression



$$\hat{\mathbf{y}} = \text{softmax}(\mathbf{o}) \quad \text{where} \quad \hat{y}_j = \frac{\exp(o_j)}{\sum_k \exp(o_k)}.$$

$$\underset{j}{\operatorname{argmax}} \hat{y}_j = \underset{j}{\operatorname{argmax}} o_j.$$

- SoftMax is a nonlinear function
- The outputs of SoftMax regression are still determined by an affine transformation of input features
- Thus, SoftMax regression is a linear model.

Loss Function:

- This loss function is commonly called the cross-entropy loss:

$$l(\mathbf{y}, \hat{\mathbf{y}}) = - \sum_{j=1}^q y_j \log \hat{y}_j.$$

- Since y is a one-hot vector of length q , the sum over all its coordinates j vanishes for all but one term.

$$y \in \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}.$$

SoftMax and Derivatives

$$\begin{aligned}l(\mathbf{y}, \hat{\mathbf{y}}) &= - \sum_{j=1}^q y_j \log \frac{\exp(o_j)}{\sum_{k=1}^q \exp(o_k)} \\&= \sum_{j=1}^q y_j \log \sum_{k=1}^q \exp(o_k) - \sum_{j=1}^q y_j o_j \\&= \log \sum_{k=1}^q \exp(o_k) - \sum_{j=1}^q y_j o_j.\end{aligned}$$

$$\partial_{o_j} l(\mathbf{y}, \hat{\mathbf{y}}) = \frac{\exp(o_j)}{\sum_{k=1}^q \exp(o_k)} - y_j = \text{softmax}(\mathbf{o})_j - y_j.$$

References and Further Reading

- *FUNDAMENTALS OF MACHINE LEARNING FOR PREDICTIVE DATA ANALYTICS. Algorithms, Worked Examples, and Case Studies* John by D. Kelleher Brian Mac Namee Aoife D'Arcy (Permission taken MIT Press)
- *Dive into Deep Learning*, Aston Zhang, Zachary C. Lipton, Mu Li, and Alexander J. Smola (July 25, 2021)
- [Intro to logistic regression: jkan-43-154.pdf \(koreamed.org\)](http://jkan-43-154.pdf)